

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug



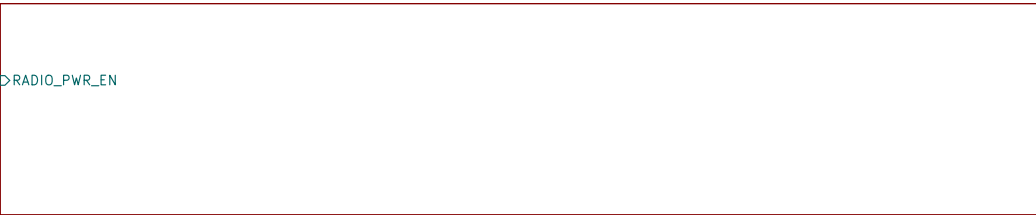
File: mcu_clocks_debug.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

ESP32-C6 radio



File: radio_esp32.kicad_sch

RADIO_PWR_EN

Sheet: /

File: ereader_rev1.kicad_sch

Title: RA8P1 e-reader - schematic index

Size: A3

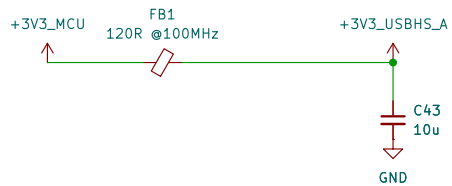
Date: 2026-09-05

Rev: 0.3 DRAFT

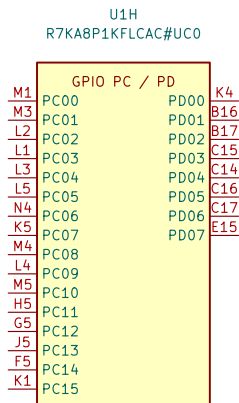
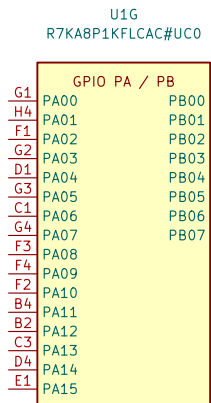
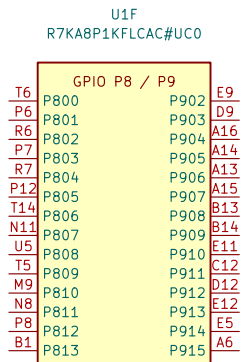
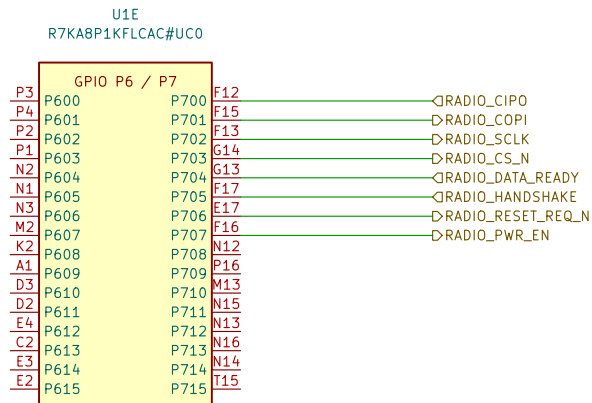
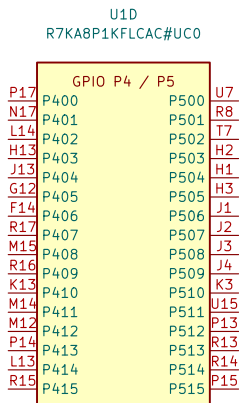
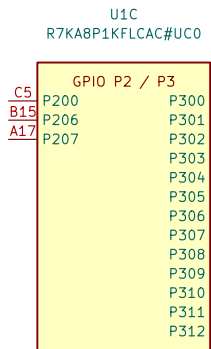
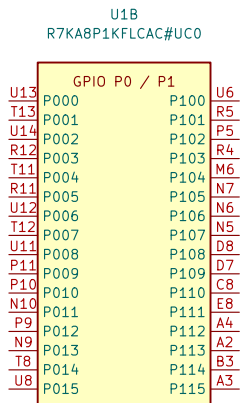
KiCad E.D.A. 10.0.5

Id: 1/6

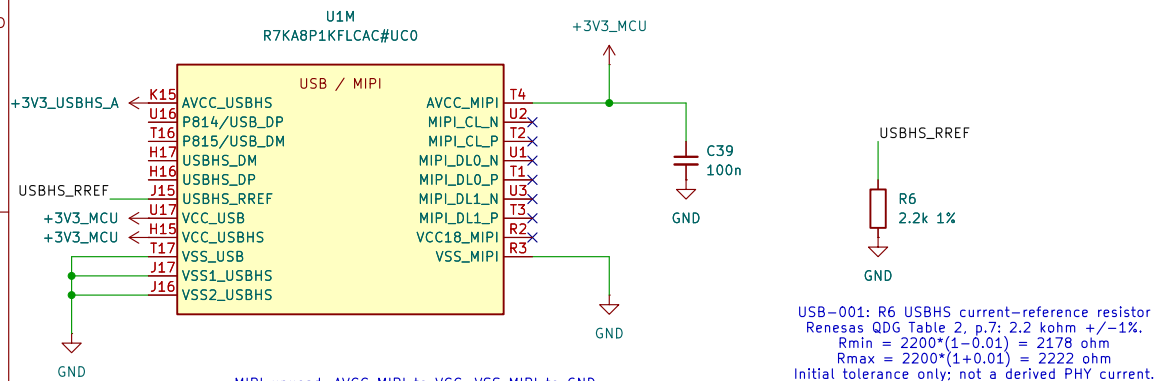
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]



RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation



MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90.110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

USB-002: USB full-speed supply / U1.U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90.110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md

USB-003: USB-HS supply / U1.H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9.11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6.56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

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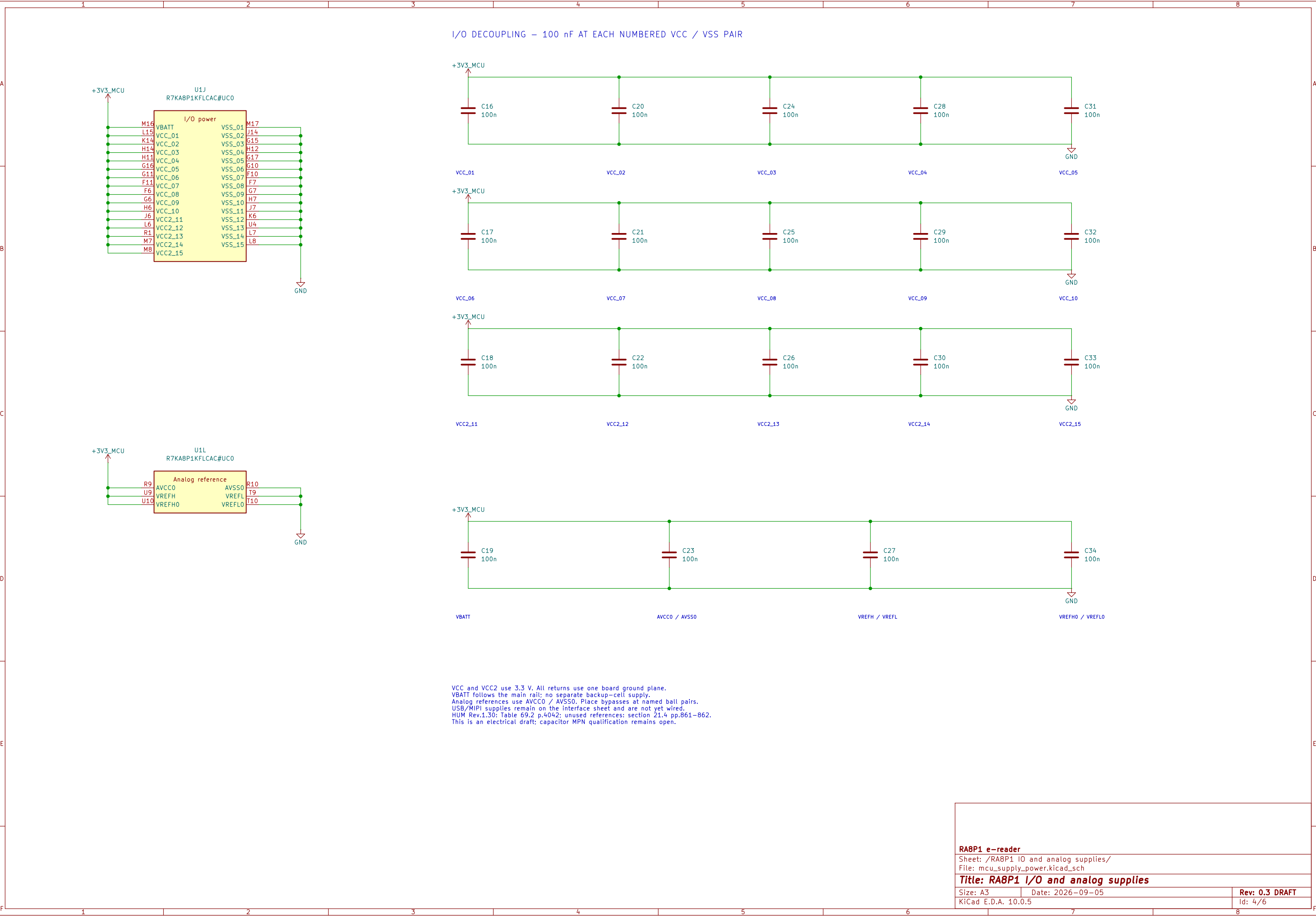
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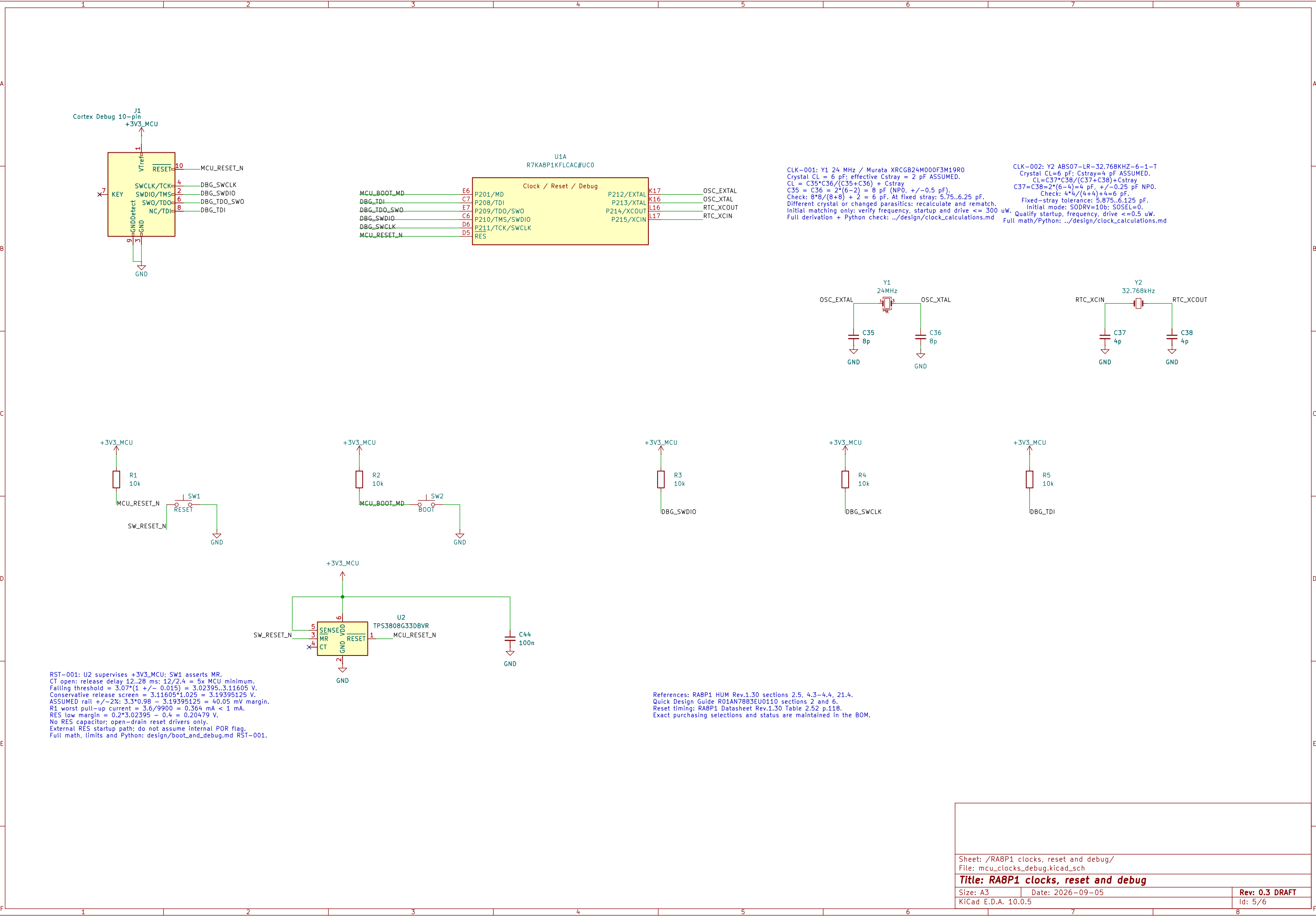
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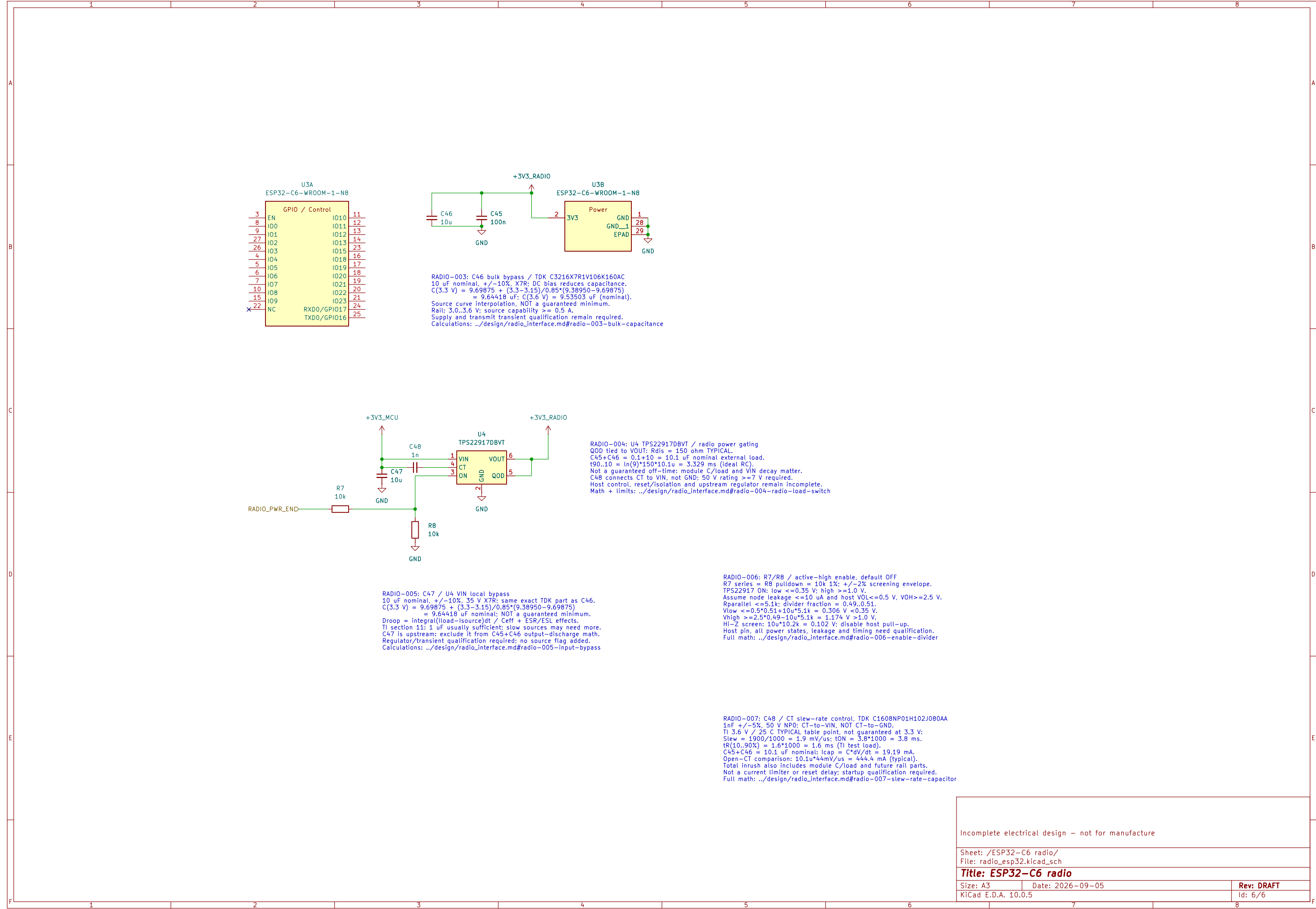
Rev: 0.3 DRAFT

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Id: 2/6







RADIO-005: C47 / U4 VIN local bypass
10 uF nominal, +/-10%, 35 V X7R; same exact TDK part as C46.
 $C(3.3\text{ V}) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)$
 $= 9.64418\text{ uF}$ nominal; NOT a guaranteed minimum.
Droop = $\text{Integral}(\text{Iload}-\text{Isource})dt / C_{\text{eff}} + \text{ESR/ESL effects}$.
TI section 11: 1 uF usually sufficient; slow sources may need more.
C47 is upstream: exclude it from C45+C46 output-discharge math.
Regulator/transient qualification required; no source flag added.
Calculations: ../design/radio_interface.md#radio-005-input-bypass

RADIO-006: R7/R8 / active-high enable, default OFF
R7 series = R8 pulldown = 10k 1%; +/-2% screening envelope.
TPS22917 ON: low $\leq 0.35\text{ V}$; high $\geq 1.0\text{ V}$.
Assume node leakage $\leq 10\text{ uA}$ and host VOL $\leq 0.5\text{ V}$, VOH $\geq 2.5\text{ V}$.
Rparallel $\leq 5.1\text{k}$; divider fraction = 0.49..0.51.
Vlow $\leq 0.5*0.51+10\text{u}*5.1\text{k} = 0.306\text{ V}$ $< 0.35\text{ V}$.
Vhigh $\geq 2.5*0.49-10\text{u}*5.1\text{k} = 1.174\text{ V}$ $> 1.0\text{ V}$.
HI-Z screen: $10\text{u}*10.2\text{k} = 0.102\text{ V}$; disable host pull-up.
Host pin, all power states, leakage and timing need qualification.
Full math: ../design/radio_interface.md#radio-006-enable-divider

RADIO-007: C48 / CT slew-rate control, TDK C1608NP01H102J080AA
1nF +/-5%, 50 V NP0; CT-to-VIN, NOT CT-to-GND.
TI 3.6 V / 25 C TYPICAL table point, not guaranteed at 3.3 V:
Slew = $1900/1000 = 1.9\text{ mV/us}$; tON = $3.8*1000 = 3.8\text{ ms}$.
tr(10..90%) = $1.6*1000 = 1.6\text{ ms}$ (TI test load).
 $C45+C46 = 10.1\text{ uF}$ nominal; lcap = $C*dV/dt = 19.19\text{ mA}$.
Open-CT comparison: $10.1\text{u}*44\text{mV/us} = 444.4\text{ mA}$ (typical).
Total inrush also includes module C/load and future rail parts.
Not a current limiter or reset delay; startup qualification required.
Full math: ../design/radio_interface.md#radio-007-slew-rate-capacitor

Incomplete electrical design – not for manufacture

Sheet: /ESP32-C6 radio/
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Rev: DRAFT

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Id: 6/6